Academic Year: 2025-26 (Even Semester)

Major Project Phase-I (CSEP23605)

Review 0 - Synopsis Submission

Date		**Domain**	**Secure DevOps**
Group Number	**G14**	**Project ID**	**P43**
Project Title	**Implementation of a Secure CI/CD Pipeline Using DevOps Tools**		

Problem Description:

In modern software development, security vulnerabilities are often detected only after application deployment. This delayed detection increases the risk of cyberattacks, leads to high remediation costs, and causes delays in software releases. Traditional DevOps pipelines focus mainly on speed and automation, often neglecting security integration during early development stages. Hence, there is a strong need for a Secure DevOps (DevSecOps) approach that integrates security testing and controls directly into the CI/CD pipeline to ensure early detection and mitigation of vulnerabilities.

Objectives:

1. **To design and implement a secure CI/CD pipeline**
2. **To integrate security testing tools into DevOps workflows**
3. **To identify and mitigate vulnerabilities at early stages of development**
4. **To reduce security risks and remediation costs**

Literature survey Completed So far and Gap Identified through the Literature Survey

Sl. #	Title of the paper and year	Methodology	Advantages	Disadvantages
1	Implementing and Automating Security Scanning to a DevSecOps CI/CD Pipeline , 2023	Automated DevSecOps CI/CD with integrated SAST and DAST scanning	Early vulnerability detection and faster remediation	Tool-dependent with limited real-world validation
2	Automated Security Findings Management: A Case Study in Industrial DevOps,2024.	Automated semantic knowledge base for unified security findings.	Enhanced data visibility and quality.	Manual rule definition required.

3	Challenges affecting the successful adoption of DevOps practices: A systematic literature review ,2024.	PRISMA-based systematic literature review of 21 studies	Identifies success factors and offers structured frameworks for challenges	Highlights cultural resistance, high costs, skill gaps, and legacy complexity
4	**Enhancing Smart Contract Security Through DevSecOps: An Adaptive Approach for Vulnerability Detection , 2025**	**Adaptive DevSecOps with AI and testing tools**	**Early vulnerability detection**	**Complex to implement**
5	Cybersecurity in DevOps Environments: A Systematic Literature Review ,2025.	Systematic literature review.	Broad evidence and taxonomy.	No empirical validation.
6	Effort Estimation in Agile and DevOps: A Systematic Literature Review of Stakeholder Dissatisfaction ,2025	Systematic Literature Review using PICOS & PRISMA	Identifies causes of estimation failure	No new estimation model
7	Extensive Review of Threat Models for DevSecOps ,2025	Systematic review of threat models with layered DevSecOps framework for single and multicloud	Comprehensive threat model coverage with early security integration support	Conceptual study without experimental validation
8	Empirical Investigation of Key Enablers for Secure DevOps Practices, 2025	Empirical Study / Survey-based Research.	Statistically validated results, industry support, sustainability focus	No technical implementation, survey bias risk, no performance benchmarking
9	Building Resilient CI/CD Pipelines: A DevOps Security-First Framework ,2025	Case study implementing a secure CI/CD pipeline using AWS and DevOps tools.	Improves security, automation, and deployment reliability.	Complex setup and limited generalization.
10	Enhancing DevOps Continuous Monitoring Phase: Hybrid Intrusion Detection and Ensemble Learning System (HIDELS),2026	Experimental / Machine Learning based	High accuracy, strong robustness, DevSecOps automation, lower bias.	High cost, no deep learning, limited scalability discussion

Panel Members Remarks (if any):	
Status:	○ **Approved** ○ **Rejected** ○ **Revision needed**

	Name	**Signature**
Panel Member 1		
Panel Member 2		
Project Guide	**Prof. Sameena H S**	
Project Coordinator	**Dr. Shruthi G**	
HOD, CSE	**Dr. R Rajkumar**	